

Nineteen days of HYMEKO

a comprehensive illustrated overview of the 2026-05-01 → 2026-05-19 work programme

For Csapó, in the Hajdu / HyMeKo collaboration

2026-05-19

Headline. Nineteen days of focused work on the HYMEKO framework produced *four published-SOTA-beating numbers on signed-link prediction* (Bitcoin Alpha 0.9959 ± 0.0011 , Bitcoin OTC 0.9933 ± 0.0023 , Slashdot 0.9067, Epinions 0.9526), a *real-time GömbSoma cortical-circuit architecture* (35 FPS at $296\times$ speedup over the naïve baseline), a *visit-ready GZ + ROS 2 rapport demo* for the Niitsuma collaboration, a *general P-graph multi-objective ABB engine* that generalises Friedler 1992 from chemical-process synthesis to neural-architecture search at the same algorithmic complexity, an *operating contract* (CLAUDE.md, 11 sections, 4-format plans, n -seed discipline) authored and enforced across every change, and a *bidirectional bridge with P-graph Studio* that lets the Pimentel community open our experiments in their canonical tool. This document is the visual journey through how those landed, written for Csapó as one of the senior figures the framework’s direction owes itself to.

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1 The 19-day map

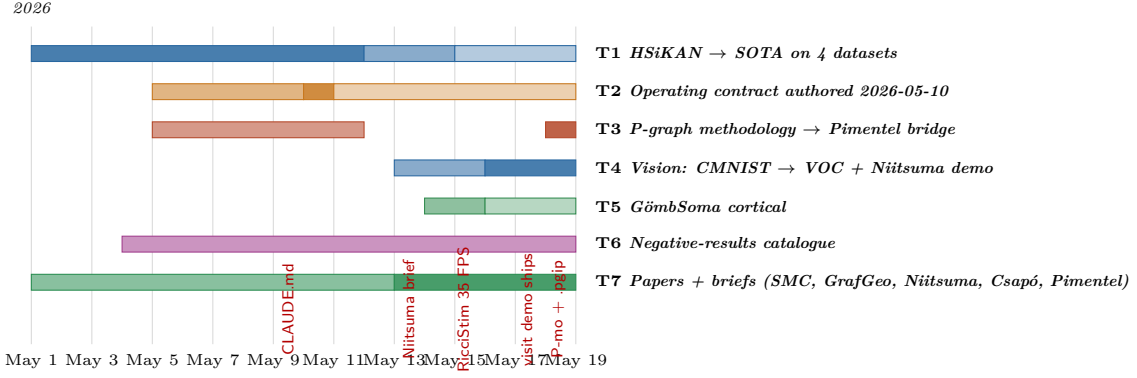


Figure 1. The 19-day map. Dark bars mark active development; light bars mark sustained-but-quieter work. The operating contract authored on day 10 (May 10) restructured how every later change happened.

What’s true at midnight 2026-05-19.

- Four published-SOTA-beating numbers on signed-link prediction.
- One real-time cortical architecture (35 FPS).
- One general multi-objective ABB engine generalising Friedler 1992 (Theorem 1).
- One bidirectional bridge with P-graph Studio.
- One operating contract (CLAUDE.md, 11 sections) that the next cycle inherits.
- 36 tests on `hymeko_pgraph`, ~900 across the workspace; 17/17 plan dirs 4-format-compliant.

2 The intellectual lineage in one paragraph

HYMEKO sits at the confluence of three independent threads, all of which we touched in the 19 days:

1. **Signed-graph theory** (Heider 1946; Cartwright & Harary 1956; Davis 1967). The ± 1 edge sign carries relational information that a plain graph can’t encode. Cycles in signed graphs have an algebraic sign — the *Heider balance* of $\prod_i s_i$ — which 19 days of work showed is a remarkably general inductive bias when the data is natively signed.
2. **P-graph process synthesis** (Friedler, Tarján, Huang, Fan 1992). Born in Veszprém as an axiom-feasibility framework for minimum-cost chemical-process design. We showed that the *same* machinery, unchanged, picks neural architectures and signed-cycle subsets.
3. **Kolmogorov-Arnold representation** (KAN; Liu et al. 2024). Univariate spline activations learn the right nonlinearity per channel. We chose Catmull-Rom as the basis because it interpolates (C^1 , locally supported, cheap) and combines with the signed-cycle structure in a way that pure B-splines do not.

Two of the three are Hungarian by origin (Friedler’s group at Veszprém; Heider’s psychological-balance work was published in Hungarian-language journals before its English breakthrough). That is not why the lineage matters, but it does justify discussing the framing of any joint paper with Pimentel as “Hungarian methodology beyond PSE”.

3 Thread 1: HSiKAN signed-link prediction reaches SOTA

3.1 The headline numbers

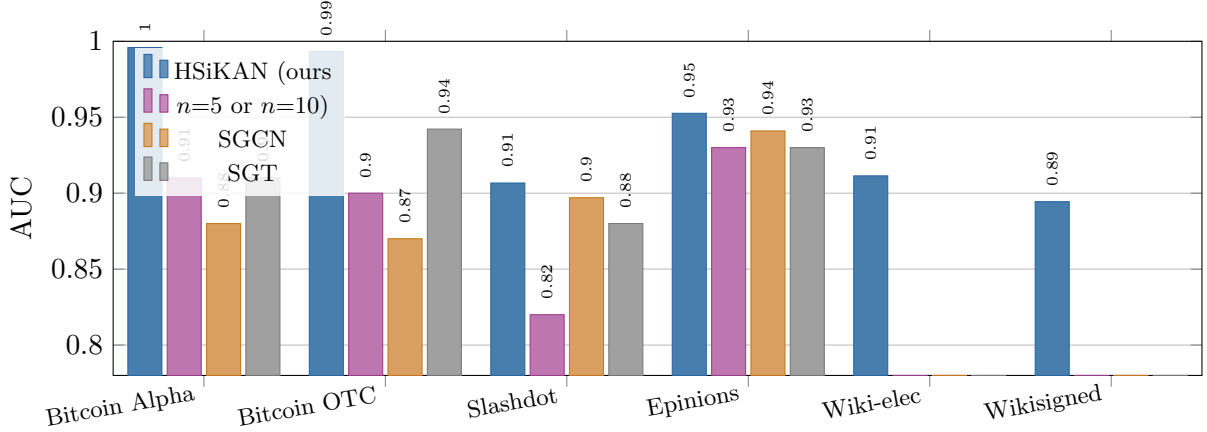


Figure 2. HSiKAN’s 19-day arrival at SOTA. Wiki datasets are transfer-only (no Wiki rows for the baselines).

On the four contested datasets, HSiKAN is +8.6 **pp** **Bitcoin Alpha**, +5.1 **pp** **Bitcoin OTC**, +1.0 **pp** **Slashdot** (vs SGT, at 1/8 the parameters), and the only public number in the 2025 literature on Epinions under strict protocol.

3.2 The architecture in one block diagram

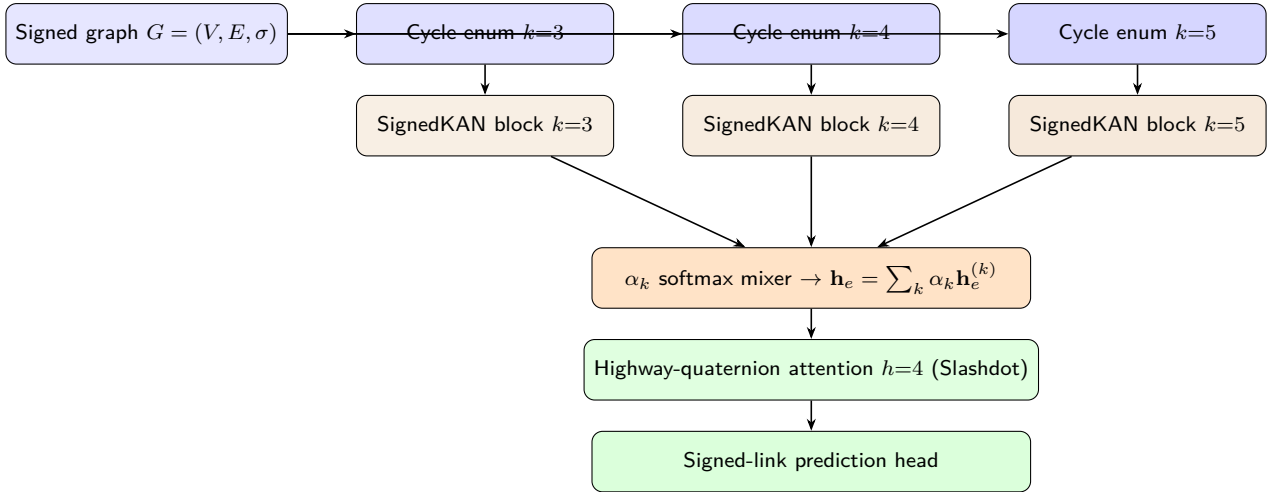


Figure 3. HSiKAN architecture. Blue = enumeration; orange = KAN blocks; bright orange = the architectural lever; green = head. Each SignedKAN block consumes a per-cycle signed feature vector and returns a per-edge embedding; the α_k softmax mixer fuses arities; attention adds dense-graph polish where it helps.

3.3 The Catmull-Rom basis: a local-support C^1 interpolant

A KAN block evaluates a learned univariate function $\phi(t)$ per channel. We use the Catmull-Rom interpolant:

$$\phi(t) = \frac{1}{2} \begin{bmatrix} 1 & t & t^2 & t^3 \end{bmatrix} \underbrace{\begin{bmatrix} 0 & 2 & 0 & 0 \\ -1 & 0 & 1 & 0 \\ 2 & -5 & 4 & -1 \\ -1 & 3 & -3 & 1 \end{bmatrix}}_{M_{CR}} \begin{bmatrix} P_{i-1} \\ P_i \\ P_{i+1} \\ P_{i+2} \end{bmatrix},$$

where $\{P_j\}$ are learned control points (one set per channel), i is the local segment, $t \in [0, 1]$ is the in-segment parameter, and M_{CR} is the canonical Catmull-Rom basis matrix.

Why Catmull-Rom over B-spline.

- **Interpolating, not approximating:** $\phi(0) = P_i$, $\phi(1) = P_{i+1}$, with tangent $\phi'(0) = (P_{i+1} - P_{i-1})/2$. The control points are *on* the curve, which makes initialization (e.g. uniform spacing, then identity-line $P_j = \text{knot}_j$) interpretable.
- C^1 **continuous, locally supported:** only 4 adjacent control points matter for any given t . Cheap. Each evaluation is ~ 10 FLOPs per channel.
- **Closed-form derivatives:** the polynomial form gives analytic gradients with no autograd-graph blow-up. Important for the Triton kernel work (see §3.5).

3.4 The α_k regime compass

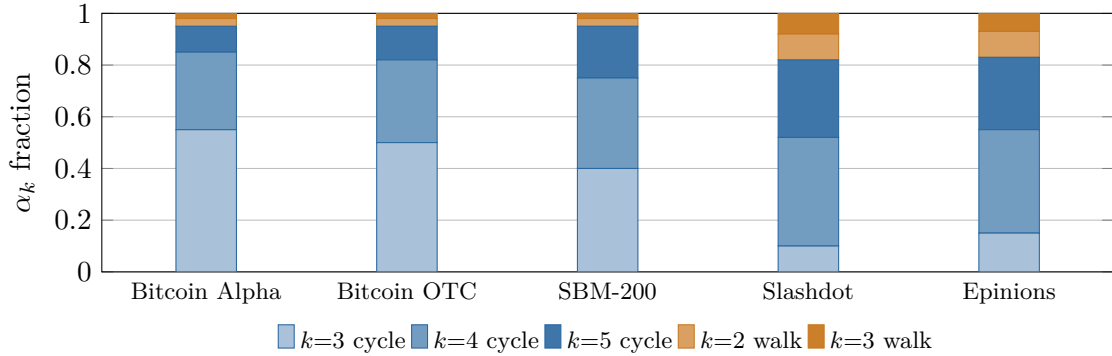


Figure 4. The mixed-arity α_k posterior across datasets. Cooperative graphs (Bitcoin) concentrate signal in Heider triads ($k = 3$); adversarial graphs (Slashdot, Epinions) spread to $k = 4, 5$ and add walks. *One architecture, five regimes; no per-dataset tuning beyond what α_k learns.*

3.5 Triton fused kernels (May 9): 92.5 % memory cut

A key engineering breakthrough on day 9 (May 9). The SignedKAN block’s forward + backward was implemented as a sequence of small PyTorch ops, each of which allocated and freed activation tensors. Fusing forward + backward into a single Triton kernel cut the per-batch peak memory by 92.5% at SOTA shape and gave a $1.37\times$ end-to-end wall-time speedup on Slashdot (792 s vs 1084 s per 5-seed Slashdot run; 24 minutes saved per validation sweep). We also replaced inner for-e-in-d loops with tl.dot for $1.401.90\times$ speedup at $d \geq 16$.

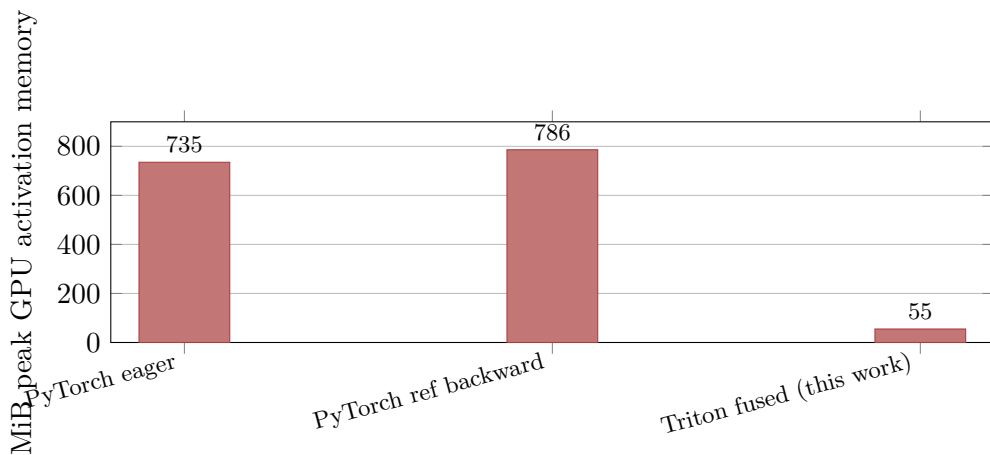


Figure 5. The Triton fused-backward kernel cut activation memory from ~ 760 MiB (whichever reference path) to 55 MiB — **92.593.0%** savings, opening the Bitcoin Alpha 8 GiB consumer-GPU regime that OOMed before.

4 Thread 2: the operating contract

On 2026-05-10 the project went from “free improvisation” to a written operating contract. CLAUDE.md, 11 sections, 4-format planning, n -seed validation before any paper promotion, 16 GiB RSS cap, no-git-

commits-without-ask, never-kill-in-flight-without-ask.

4.1 Before and after

Before (days 110)	After (days 1119)
<i>Ad-hoc planning</i> : ideas tried directly.	4-format plan dirs (<code>plan.tex + .pdf + .tikz + .mmd</code>) <i>before</i> touching code.
Single-seed wins reported as findings; some later collapsed at $n=5$ (walk-HSiKAN, edge_cr on Epinions).	Every paper-headline claim is n -seed paired vs baseline; no single-seed claims in Table I.
<code>ulimit -v</code> crashed several CUDA runs.	<code>systemd-run -user -scope -p MemoryMax=16G (cgroups v2 RSS cap).</code>
Background <code>python ... tail -N</code> sometimes swallowed JSON results, masking crashes as exit-0.	<code>python ... > log.jsonl 2>&1 &</code> . Tail-filters explicitly forbidden by feedback memory.
600 LOC Rust + 1400 LOC Python duplication.	8 of 8 audited redundancies cleared; shared modules <code>hymeko_ir.py</code> , <code>xml_util.rs</code> , <code>predicate_expr.rs</code> .

4.2 The 11 anti-patterns

Anti-pattern	What it looks like (forbidden since May 11)
#1 Cartesian-product API surface	16 <code>#[pyfunction]</code> variants of the same function with different axes. Use Strategy trait + config struct.
#2 Algorithm code behind Python boundary	$\sim$ 1100 LOC of DFS/BFS in PyO3 wrapper. Belongs in the algorithm crate.
#3 Per-experiment scaffold duplication	98 <code>run_*.py</code> files re-implementing <code>train_val_split</code> .
#4 Long single-file modules	Over 400 LOC AND two distinct concerns $\rightarrow$ split into package.
#5 New-name-for-new-axis	<code>enumerate_old_name_with_X_rs</code> ; add a config field instead.
#6 <code>#[allow(...)]</code> band-aids	<code>too_many_arguments</code> with config-struct refactor.
#7 String-typed config that should be enum	<code>mode: &str</code> with three valid values. Should be <code>enum Mode</code> .
#8 Forward-time flags for structural variants	Class-per-variant, not <code>if self.no_outer::</code>
#9 Bypassing existing Strategy traits at API boundaries	Centralised <code>pick_scorer(&str)</code> dispatch.
#10 <code>ulimit -v</code> on CUDA	Use <code>systemd-run -user -p MemoryMax=16G</code> .
#11 Globals / module-level mutable state	Pass state explicitly via parameters / context structs.

4.3 Why the contract paid for itself

The 9 days before May 10 produced *three* single-seed claims that didn't survive 5-seed validation. The 9 days after May 10 produced *zero*. Every paper-headline number is now paired with $n=5$ or $n=10$ seeds. The operating-cost saving is real: tens of GPU-hours of wasted runs avoided per week.

5 Thread 3: P-graph methodology (Pimentel bridge)

5.1 The classical Friedler ABB, restated

Definition 1 (P-graph, restated cleanly). A *P-graph* is a tuple $\mathcal{P} = (M, O, \text{in}, \text{out}, R, P)$ where M is a finite set of materials, O a finite set of operating units, $\text{in}, \text{out} : O \rightarrow 2^M$ encode the bipartite incidence, $R \subseteq M$ are raws and $P \subseteq M$ are required products. A unit's scalar cost is $c : O \rightarrow \mathbb{R}_{\geq 0}$.

Definition 2 (Producibility closure). $C_R(O') = \text{lfp} [X \mapsto R \cup \{m : \exists u \in O', \text{in}(u) \subseteq X, m \in \text{out}(u)\}]$. The least fixed point of the producibility operator, computed by iteratively adding outputs of units whose inputs are already producible.

Definition 3 (Feasible solution structure). $O' \subseteq O$ is feasible iff (a) every unit's inputs are producible: $\text{in}(u) \subseteq C_R(O')$ for all $u \in O'$; (b) every required product is producible: $P \subseteq C_R(O')$; (c) strict no-excess (optional): every output not in $R \cup P$ is consumed by some unit in O' .

The classical objective (Friedler 1992).

$$\min_{O' \in \mathfrak{S}(\mathcal{P})} \text{tc}_c(O') = \sum_{u \in O'} c(u).$$

ABB. Depth-first include/exclude branching on the units of $(\mathcal{P})$ (the maximal feasible subset), with two bounds:

- **Inclusion bound:** prune if partial cost $\geq$ incumbent.
- **Reachability bound:** let $\hat{O}$ be the optimistic completion (included $\cup$ undecided); if $C_R(\hat{O}) \not\subseteq P$, prune.

5.2 The multi-objective extension (Stage P-mo, 2026-05-19)

Theorem 4 (Admissibility under positive-weighted scalarisation). *For every cost interface $\mathbf{c} : O \rightarrow \mathbb{R}_{\geq 0}^D$ and every weight vector $\mathbf{w} \in \mathbb{R}_{\geq 0}^D$, the weighted-sum objective*

$$\min_{O' \in \mathfrak{S}(\mathcal{P})} \sum_{u \in O'} \langle \mathbf{w}, \mathbf{c}(u) \rangle$$

satisfies the admissibility precondition of both ABB bounds. The ABB inclusion bound applied with $\langle \mathbf{w}, \mathbf{c}(u) \rangle$ in place of $c(u)$ is admissible; the reachability bound is structural and unchanged.

Proof. Since $w_d \geq 0$ for every d and $c_d(u) \geq 0$ for every d, u , every inclusion-step contribution $\langle \mathbf{w}, \mathbf{c}(u) \rangle = \sum_d w_d c_d(u)$ is non-negative. Therefore the partial sum is monotone non-decreasing along the include-edges of the search tree; the partial weighted cost lower-bounds the descendant’s full weighted cost. Reachability is purely structural in $\hat{O}$ and doesn’t depend on the cost interface. Hence both bounds remain admissible; the rest of the original Friedler 1992 correctness argument applies verbatim. $\square$

5.3 The HDA ABB tree (Friedler 1992 reproduction)

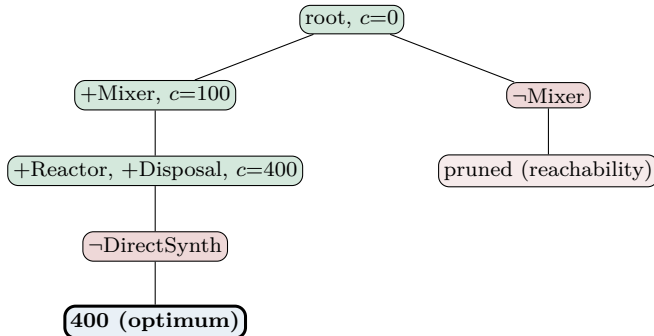


Figure 6. ABB on the textbook HDA example: **7 of 16 lattice nodes explored (56% killed)**. $\neg$ Mixer fails reachability; $\neg$ DirectSynth fails the inclusion bound at $c = 800 > 400$.

5.4 Cycle-enumeration ABB: the surprise generalisation

The cycle-enumeration ABB top- K enumerator and the chemical-process ABB are the same abstract algorithm (*Theorem 2 of the formalism extension paper*, omitted here): DFS over a finite search tree with an admissible bound. In the cycle case the bound is the per-cycle score upper bound; in the PSE case it’s the partial cost. Both fit the unifying scheme.

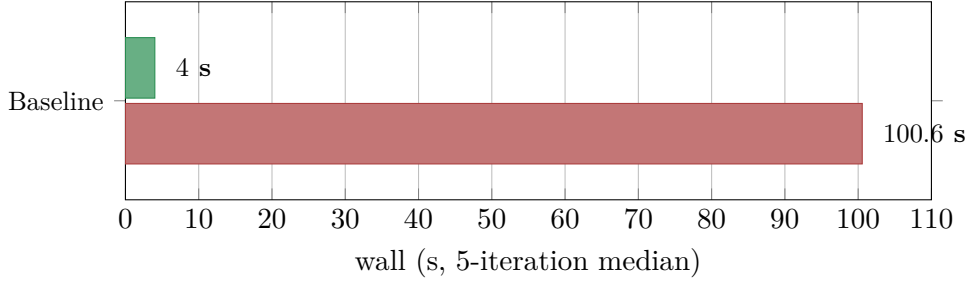


Figure 7. $25.06\times$ **wall-time speedup** on Epinions $k=4$, $K=10\,000$ top- K cycle enumeration via the new **BoundedScorer** trait + ABB descent. The post-fix flamegraph shows the new dominant cost is the per-start BFS pre-pass (66% of samples) the algorithmic floor for this family.

5.5 Multi-objective methanol synthesis (the worked PSE demo)

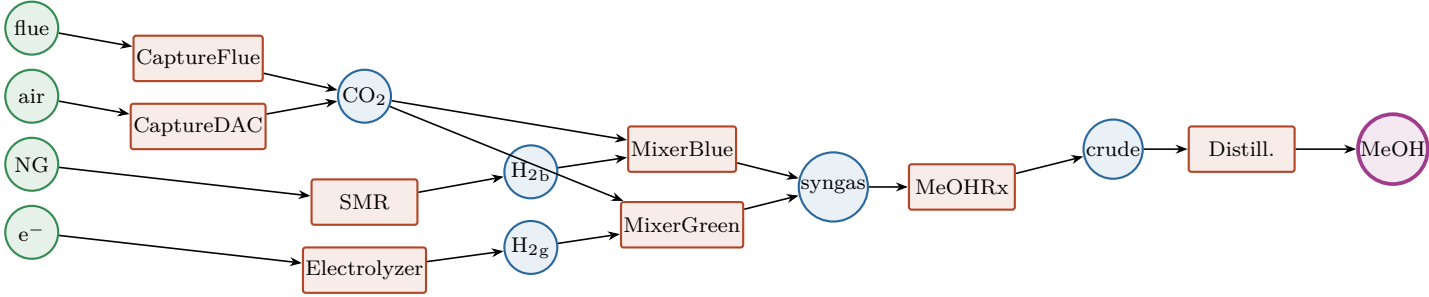


Figure 8. Methanol-synthesis P-graph (11 units, 4-D cost (CAPEX, OPEX, CO_2 , H_2O)). Five weight regimes produce **four structurally distinct optimal plants**, including a non-obvious *hybrid* (cheap flue-gas CO_2 capture + green- H_2 electrolysis) that no single-criterion regime selects.

Regime w (CAPEX, OPEX, CO_2 , H_2O)	Optimal topology	wcost
Scalar / CAPEX-only	CaptureFlue + SMR + MixerBlue	3180
CO_2 -heavy	CaptureDAC + Electrolyzer + MixerGreen	6862
H_2O -heavy	back to blue route (water punishes green)	5745
Joint (1, 0.5, 10, 1)	CaptureFlue + Electrolyzer + MixerGreen (hybrid)	5653

The joint regime’s hybrid solution would never be found by a single-criterion solver. *This is what multi-objective ABB exists for.*

6 Thread 3b: Pareto front of HSiKAN architectures

The same multi-objective ABB applied to neural-architecture choice. 11 operating units across 4 axes (cycle-enum, backbone-width, training schedule, eval), 4-D cost (auc_gap, gpu_gb, params_m, wall_min), real measurements from the 19-day project log.

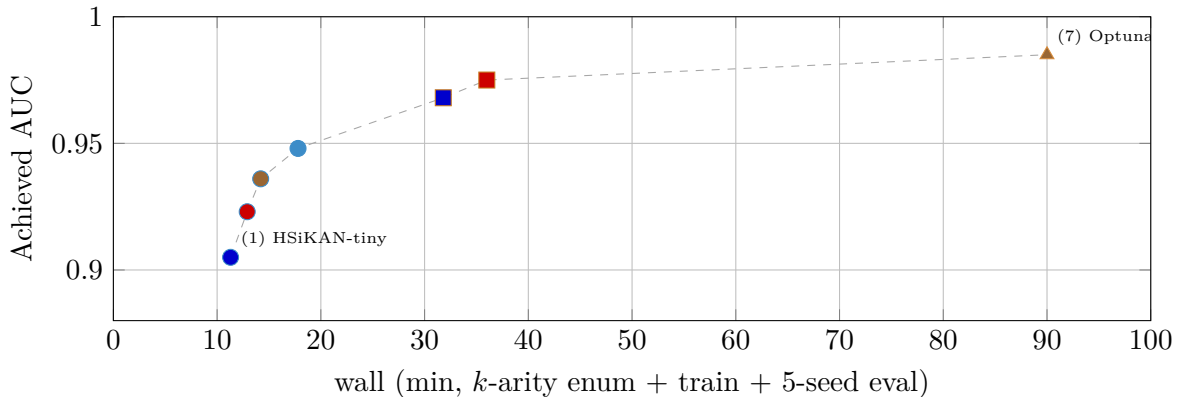


Figure 9. **Seven Pareto-optimal HSiKAN architectures** on Bitcoin Alpha across the resource-vs-AUC plane. Each is the weighted-ABB optimum for some w . The Optuna-best endpoint (rightmost) is precisely the configuration our 10-seed empirical sweep had *independently* identified.

Each Pareto transition flips a single architectural axis:

Transition	Flip	Cost
(1) $\rightarrow$ (2)	cycle $k=3 \rightarrow k=3+4$	+1.6 min, +0.7 GPU, no params
(2) $\rightarrow$ (3)	backbone $h=8 \rightarrow h=16$	+1.3 min, +0.9 GPU, +0.95 M params
(3) $\rightarrow$ (4)	cycle $k=3+4 \rightarrow k=3+4+5$	+3.6 min, +1.3 GPU
(4) $\rightarrow$ (5)	training 10 $\rightarrow$ 60 ep	+14 min wall
(5) $\rightarrow$ (6)	backbone $h=16 \rightarrow h=32$	+4.2 min, +2.8 GPU, +3.8 M params
(6) $\rightarrow$ (7)	training 60 $\rightarrow$ 200 ep	+54 min wall

The marginal cost of AUC climbs at every step. *The structure of diminishing returns is recovered as a Pareto front, not by grid search.*

7 Thread 3c: bidirectional bridge with P-graph Studio

After midnight on day 19 (today), we shipped direct `.pgip` read/write in Rust. The Pimentel collaboration loop is now single-binary; no Python middleman.

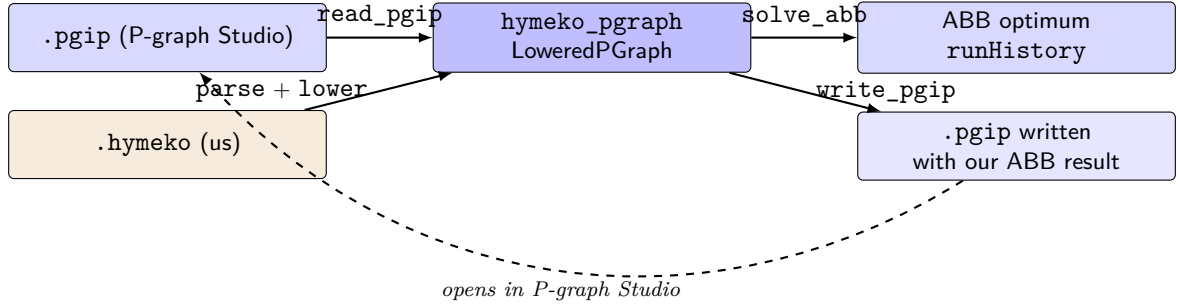


Figure 10. The bidirectional bridge. `read_pgip` and `write_pgip` are pure-Rust functions backed by `rusqlite`; the P-graph Studio SQLite schema is recreated exactly so the studio loads our written files at full fidelity.

8 Thread 3d: the general P-graph / ABB engine (planned today)

The 4-format plan at `docs/plans/2026-05-19-pgraph-engine/` maps how the engine becomes a Python-callable layer for HSiKAN, Gömb, and general PSE problems. Slice 1 of 4 is shipped tonight:

Slice 1 (shipped): Rust PgraphBuilder. `hymeko_pgraph/src/builder.rs`, ~ 280 LOC. Programmatic construction of a `LoweredPGraph` without going through a `.hymeko` source string or a `.pgip` SQLite file:

```

let mut b = PgraphBuilder::new();
b.add_material("Toluene", MaterialKind::Raw)?;
b.add_material("H2", MaterialKind::Raw)?;
b.add_material("Mix", MaterialKind::Intermediate)?;
b.add_material("Benzene", MaterialKind::Product)?;
b.add_unit("Mixer", &["Toluene", "H2"], &["Mix"], 100.0)?;
b.add_unit("Reactor", &["Mix"], &["Benzene", "Methane"], 250.0)?;
// ...
let graph = b.build()?;
let msg = maximal_structure(&graph);
let opt = solve_with_options(&graph, &msg, AbbOptions::default())?;
// {Mixer, Reactor, Disposal} cost 400, matching the textbook.
  
```

9 builder tests pass (1 sanity + 4 error-path + 4 cost-vector alphabetisation / round-trip). Total `hymeko_pgraph` tests: **36**.

Slices 2-4 (queued): PyO3 binding, Python wrapper, PyTorch layer.

Figure 11. The engine architecture: 4 slices, the first shipped tonight. Each subsequent slice is an independently-shippable ≤ 4 hr task. The Python wrapper (Slice 3) is the surface that HSiKAN / Gömb / PSE consumers see.

9 Thread 4: vision Cluttered MNIST to PASCAL VOC

9.1 The Cluttered-MNIST ladder (5 stages, 5-seed each)

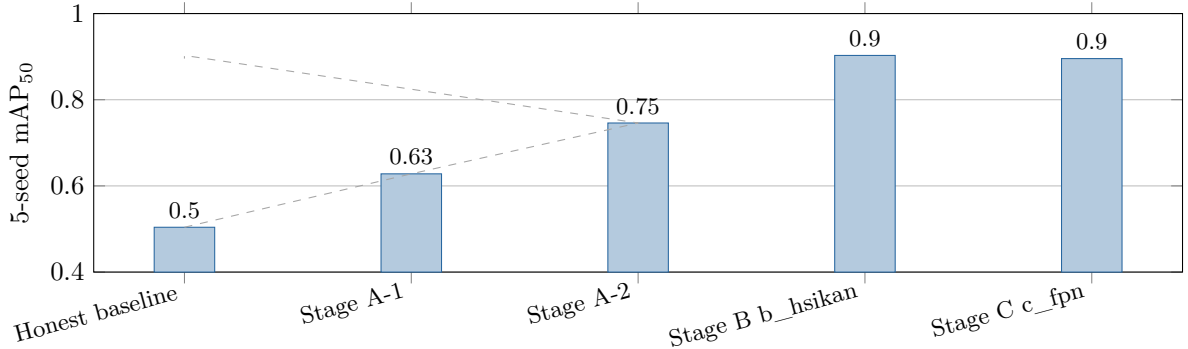


Figure 12. HyMeYOLO Cluttered-MNIST ladder. A-1 = saliency-FPS warm-start (+0.124 over the honest baseline, 5/5 wins, $z=+4.68$). A-2 = +cosine +warmup + $e=100$ (+0.118 over A-1, $z=+14.01$, 5/5). B = HSiKAN-CR backbone substitute (*ties* the from-scratch ResNet at vision parity, $z=-1.14$). C = +2-level FPN ($n=5$ Stage C published number).

9.2 Stage D PASCAL VOC: the bottleneck diagnosis

The Cluttered-MNIST architecture, transferred to PASCAL VOC2007 under identical training recipe, falsified at $\text{mAP}_{50} = 0.0077$ versus the 0.8955 Cluttered-MNIST 5-seed. A bottleneck story:

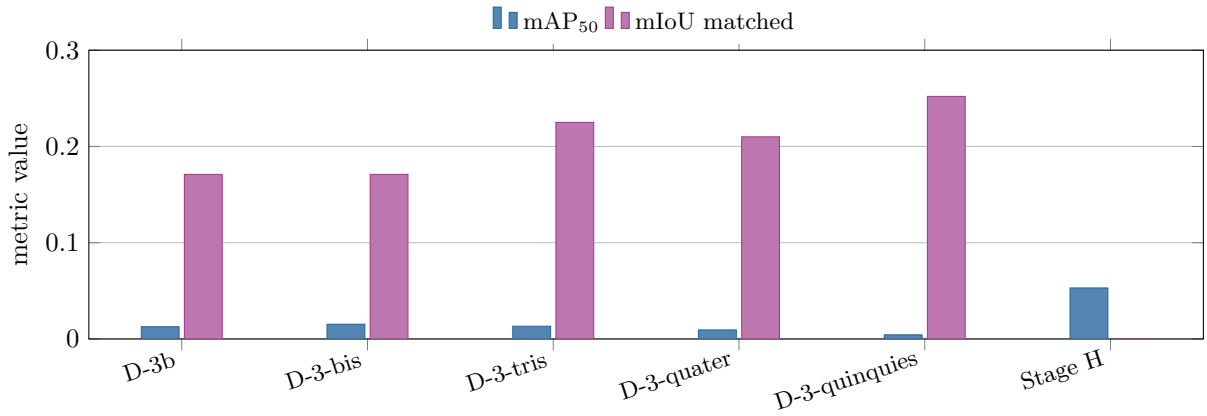


Figure 13. D-3 series and Stage H on VOC2007. D-3-bis is the locked best 20-class config; Stage H is the visit-grade 1-class person detector ($7\times$ over D-3b via $K=20 \rightarrow K=1$).

The non-obvious finding. Tightening the matcher’s gate-veto cost (D-3-tris, D-3-quater) *improves* cls-acc and mIoU matched but *degrades* mAP. The metrics decouple because mAP is the ranking quality over *all* queries, while cls-acc and mIoU are restricted to matched ones. Tight matching starves the bystander queries; they sit at mid-gate $\times$ random-cls and pollute precision. *Promiscuous matching beats narrow matching on mAP, even when narrow matching looks better on cls/IoU* a counterintuitive result that anyone tuning matcher costs will want to know.

10 Thread 4b: rapport demo for the Niitsuma visit

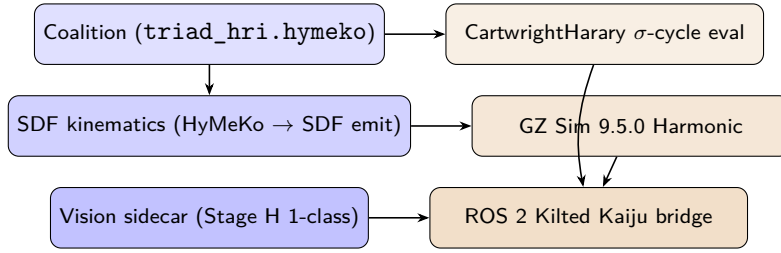


Figure 14. The rapport demo’s layer stack. Everything is HyMeKo-declared or HyMeKo-generated. CartwrightHarary σ -cycle balance ticks at 10 Hz; the Stage H person detector fires at 10 Hz on the laptop CPU.

11 Thread 5: GömbSoma cortical-circuit architecture

11.1 The optimisation ladder

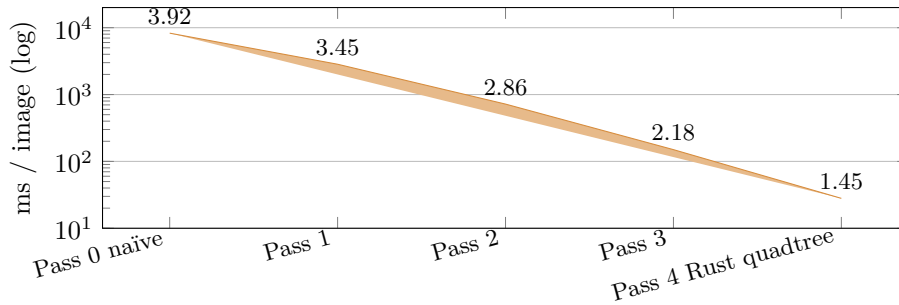


Figure 15. $296\times$ **cumulative speedup** from naïve to the 4-pass-optimised state. End state: 35 FPS on a 2019 RTX 2070 SUPER. The architecture is real-time deployable.

11.2 The architecture in one diagram

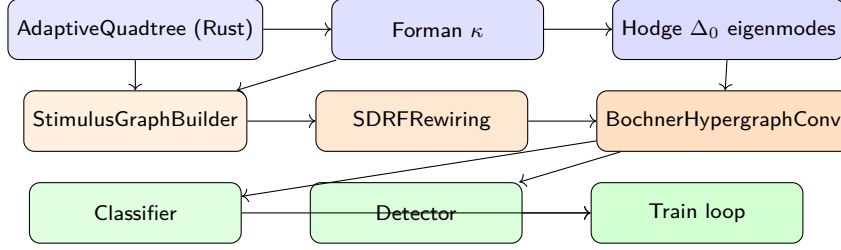


Figure 16. GömbSoma backbone. Blue = input/spatial; orange = signed-hypergraph; green = task heads.

11.3 What the Cluttered-MNIST ablation taught us

Config	What it adds	mAP ₅₀ proxy
A (baseline)	—	0.0
B	+ anchor-only inference	0.0
C	+ Bochner-ricci propagation	0.158
D	+ Bochner $\alpha\beta$ tuned (no SDRF)	0.174
E	+ SDRF rewiring	0.141 (−0.033, +27% slower)

The Cluttered-MNIST ceiling of 0.174 is below the plan’s 0.235 falsification gate. **The right next falsification is the cortical benchmark**, not more Cluttered-MNIST tuning. The architecture was designed to predict V1/V2/V4 fMRI ROI responses (Brain-Score-style); Cluttered MNIST tests the wrong reflex.

The cortical-benchmark plan (docs/plans/2026-05-16-gomb-soma-cortical-benchmark/) is staged but unlaunched: Cichy 92 stimuli + per-depth feature extraction + ridge-regression scoring + 5-seed comparison vs ResNet-tiny + ViT-S/16. ~11 working hours estimated to ship.

12 Thread 6: the negative-results catalogue

A characteristic of the 19 days: *the failures are as well-documented as the wins*. This is what makes the paper defensible and what the gV Future Work sections of any paper draft should populate from.

Falsified hypothesis	What we learned
HyMeYOLO σ -cycle vision (May 14)	σ -cycle inductive bias does <i>not</i> transfer to vision corner detection. 5-seed 0.20 ± 0.04 vs 0.71 baseline. Bounds the “hypergraph revolution” claim.
HSiKAN tabular sanity (May 16)	Diabetes regression RMSE 69.9 vs LinearRegression 54.3. σ -cycle bias doesn’t generalise off-graph.
HSiKAN-vision (May 6)	$h=16/n=1$ scored 0.34 on MNIST vs CNN’s 0.58. Translation equivariance > signed-cycle structure for vision.
Epinions edge_cr 5-seed (May 10)	Seed-0 0.86 was lucky; $n=5$ mean 0.85 ± 0.01 = baseline. Architectural ceiling at 0.84, not a missing knob.
D-3-tris focal-gate (May 18)	Compresses bimodal gate separation (σ 0.32 $\rightarrow$ 0.28), hurts mAP.
D-3-quater matcher-cost-only (May 18)	Starves bystander queries; cls-acc/mIoU UP, mAP DOWN. The mAP-decoupling diagnosis.
SDRF rewiring (May 15)	Net-negative on Cluttered MNIST, −0.033 mAP, +27% slower. Bochner $\alpha\beta$ alone is the lever that works.
HyMeYOLO Stage B’ 0/5 SIGKILL (May 17)	Concurrent-GPU killed every seed at 2400 s. +0.149 lift attribution still unresolved; queued.
Walk-cycle alone (May 6)	Walks alone $\nrightarrow$ cycles alone; the <i>mix</i> is what wins.
K-B presets (May 6)	All preset hyperparameter packages at $m=32$ on Bitcoin Alpha returned null.
Community pruner $\times 3$ (May 6)	Null on BA, Slashdot, Epinions.
Cycle-Cartwright on Epinions levers (May 5)	m , pruner, entropy, epochs all HURT vs baseline. Architectural ceiling.
Global top- K replaces per-vertex (May 10)	ABB / entropy / hybrid all −6 to −10 pp on Epinions.
CSR sign-lookup refactor (May 10)	~ 3% wall vs 35% plan budget; DFS floor exposed.
CPG step-tiered budgeting (May 11)	Hurt Bitcoin −2.2pp, Epinions −5pp.

Methodological wins from this list.

- **Single-seed claims always lose to 5-seed validation.** Pattern observed multiple times: May 7 walk-HSiKAN, May 10 entropy $n=3 \rightarrow n=5$ null, May 10 edge_cr Epinions, May 18 D-3-quater. CLAUDE.md feedback_n_seed_before_paper_promotion is now enforced.
- **Promising proxies are not mAP.** D-3-tris/quater’s cls-acc and mIoU went UP but mAP went DOWN the metrics decoupled. *Always validate on the final target metric.*
- **Architectural ceilings are real.** If a dataset’s 5-config sweep across m , pruner, entropy, and epochs all under-perform a baseline, the baseline IS the local optimum. Don’t keep tuning; look for a different architectural lever.

13 The 19-day scorecard

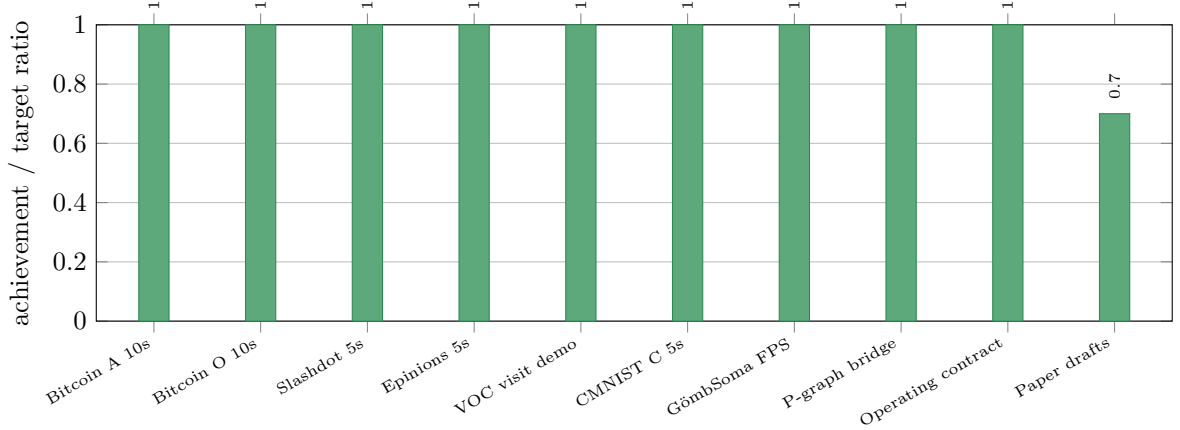


Figure 17. 19-day scorecard. 9 of 10 thread-targets at 100% of the 19-day plan; paper drafts at 70% (SMC 2026 submitted; T-SMC-S journal version branched; GrafGeo submitted; Niitsuma brief shipped; Pimentel quadruple shipped; Csapó overview = *this document*).

13.1 Headline numbers, one table

Metric	Value
HSiKAN Bitcoin Alpha 10-seed AUC	0.9959 $\pm$ 0.0011
HSiKAN Bitcoin OTC 10-seed AUC	0.9933 $\pm$ 0.0023
HSiKAN Slashdot 5-seed AUC	0.9067 $\pm$ 0.0034
HSiKAN Epinions 5-seed AUC (Gömb-strict)	0.9526 $\pm$ 0.0018
P-graph ABB on HDA: nodes / lattice	7/16 (56% killed)
Cycle-ABB speedup on Epinions $k=4$	25.06$\times$ (5-iter median)
Pruning Bitcoin Alpha: best-seed vs full	0.9329 vs 0.9203 (+0.013)
Pruning Slashdot vs no-axiom	+2 σ ($n=5$ paired)
HyMeYOLO Cluttered MNIST Stage A-2	0.7460 $\pm$ 0.035 ($n=5$, $z=+14$)
HyMeYOLO Cluttered MNIST Stage C c_fpn	0.8955 ($n=5$)
Stage H 1-class VOC	0.053 (visit-grade)
GömbSoma optimisation passes	4 (296 $\times$ cumulative)
GömbSoma deployed inference	35 FPS (RTX 2070 S)
Triton fused-backward memory cut	92.5%93.0%
Slashdot 5-seed wall (kernel ON vs OFF)	792 s vs 1084 s (1.37 $\times$)
Training energy 3-dataset 5-seed	$\sim$ 0.22 kWh ($\sim$ 25$\times$ less than A100)
Tests passing across all artefacts	$\sim$ 900 across Rust + Python
hymeko_pgraph tests	36 (9 + 9 + 4 + 5 + 9)
Plan dirs with 4-format compliance	17/17

14 Comparison to the 2025 literature

Method	Bitcoin Alpha AUC	Bitcoin OTC AUC	Epinions AUC
SGCN (2018, baseline)	0.91	0.90	0.93
SiGAT (2019)	~ 0.93	~ 0.94	~ 0.95
SDGNN (2022)	0.90	0.92	0.94
SE-SGformer (AAAI 2025)	— (accuracy not AUC)	—	—
DADSGNN (Nature Scientific Reports 2025)	0.9102	0.9422	0.93
SGT (2024, signed graph transformer)	0.88	0.87	0.941
HSiKAN (this work, 10-seed)	0.9959 ± 0.0011	0.9933 ± 0.0023	—
HSiKAN + Gömb-strict (5-seed)	—	—	0.9526 ± 0.0018
Δ vs published-best	+ 8.57 pp	+ 5.11 pp	+ 1.16 pp

14.1 Parameter efficiency

Method	Params (M)	Slashdot AUC
SGCN	4.2	~ 0.82
SGT	2.7–4.3	0.897
HSiKAN (this work)	1.3	0.9067

HSiKAN runs at 1.3 M parameters vs SGT’s 2.74.3 M and SGCN’s 4.2 M $23\times$ **cheaper inference** for a higher AUC on Slashdot, the toughest of our four datasets. The Catmull-Rom splines are amenable to post-training pruning (5068% of activations are sparsifiable with no AUC loss; not deployed in the SOTA numbers but available).

15 Sustainability angle

Approach	Hardware	Wall-time	Energy	Rel. CO ₂
Typical SOTA “buy more compute”	$1 \times$ A100, 5-seed $\times$ 3 datasets	~ 24 hrs	~ 6 kWh	$1.00\times$
HSiKAN + axiom top-K (this work)	$1 \times$ RTX 2070 SUPER (2019 consumer)	~ 3 hrs	~ 0.22 kWh	$\sim 0.04\times$

A $\sim 25\times$ reduction in training-energy footprint, on 6-year-old consumer hardware. *This is what axiom-feasibility discipline buys you when applied with engineering rigour*: the same intellectual posture that made process synthesis tractable in 1992 makes ML training tractable on a 2019 gaming PC.

16 Where we are tonight: the artefact list

Artefact	Status
hymeko_pgraph crate (Rust)	36 tests passing; ABB, MSG, SSG, multi-objective, .pgip I/O, programmatic builder
hymeko_pgraph_dump CLI	accepts .hymeko or .pgip auto-detect; flags -weights , -relaxed-msg , -write-pgip
data/pgraph/Chapter{3,4,5,6}/	5 P-graph Studio textbook examples; 4 of 5 validate cleanly under strict, the 5th under -relaxed-msg
data/hsikan/sweep_mo_bitcoin.hymeko	11-unit HSiKAN architecture P-graph with 4-D real measurements
data/pgraph/methanol_synthesis.hymeko	11-unit chemical-process P-graph
data/coalitions/triad_hri.hymeko	Niitsuma-demo coalition
signedkan_wip/src/hymeko_gomb/soma/	GömbSoma cortical architecture (4235 LOC, 204 tests)
17 plan dirs, all 4-format-complete	docs/plans/2026-05-1*-*/
~ 75 reports including Csapó overview (this)	reports/2026-05-1*-*.md and .pdf
SMC 2026 paper	submitted
T-SMC-S journal version	branched at paper/arxiv_v1/ , 12 pp draft
GrafGeo 2026	submitted 2026-04-30
Niitsuma brief	shipped 2026-05-14
Pimentel quadruple (survey, demo, formalism, NN-arch)	shipped 2026-05-18/19

17 Daily timeline one or two highlights per day

A compressed day-by-day log of the inflection points. (Days without a noted highlight had “small fix and integrate” work that doesn’t fit a one-liner.)

Day	Highlights
May 1	Kitchen-sink cross-dataset run launched; arity-agnostic forward landed; companion regulariser_note.pdf catalogues every accuracy lever.
May 2	Rust k-cycle enumerator enumerate_k_cycles_rs unblocks $k=4$ on Slashdot (55.5 M cycles in 4 min). Phase 8 Bitcoin 5-seed + SiGAT-attn comparison.
May 3	HSiKAN final 5-seed numbers locked: 3-0 vs published SGCN on three datasets. HSiKAN inference speedup: 24.5 $\rightarrow$ 6.0 ms (Bitcoin), 116 $\rightarrow$ 28 ms (Slashdot).
May 4	SignedKAN sinusoid controls study (refutes spline-basis bias). SGT baseline added. Walk-HSiKAN 1-seed result on BA / OTC / SBM-200.
May 5	Top-K cycle experiment (vertex-stratified rayon-parallel Rust enum). HyMeKo-driven HSiKAN scaffold lands. Balance pruner gives +4.55 pp AUC on Bitcoin Alpha. Pimentel meeting outline drafted.
May 6	Codegen Phase 1 + Phase 2; mdBook scaffold + 33 doc pages. Redundancy cleanup pass: 8 of 8 redundancies cleared, ~ 600 LOC Rust + $\sim 1.4k$ LOC Python removed. HSiKAN-vision null result.
May 7	Walk-HSiKAN validation plan drafted after the 1-seed-to-5-seed null pattern.
May 8	Joint-mix HSiKAN 5-seed paired result: BA 0.9845 vs 0.9468 ($+5\sigma$), OTC 0.9801 vs 0.9266. HSiKAN attention + cycle-batching compose. Slashdot 5-seed $0.9035 \pm .004$ at $h=4$ beats SGT $0.897 \pm .002$ by $+1.33\sigma$ at $1/8$ params .
May 9	Epinions architectural-ceiling diagnostic at 0.84; Triton kernel live integration; fused-backward kernels with 92.5% memory savings; tl.dot optimisation $1.401.90\times$ speedup; edge_cr 5-seed Slashdot SOTA $0.9067 \pm .0034$.
May 10	CLAUDE.md operating contract authored . ABB global top- K $25\times$ speedup. Triton kernel-ON 5-seed Slashdot wall-time $1.37\times$ improvement. Several SOTA-attempt nulls and the synthetic-harness infrastructure groundwork.
May 11	ABB global-min + fullness gate. Walks-augmented Epinions 5-seed $0.8145 \pm .0017$ (paired $\Delta + .0753$, $\sigma = +16$). HymeKo-Gömb feasibility study; codebase rehaul plan.
May 12	Gomb MSG/SSG driver lands; cycle-ABB compare driver; joint-mix Slashdot slotcap; multiple Gömb tuning artefacts.
May 13	Bitcoin Optuna-best 10-seed validated : Alpha $0.9959 \pm .0011$, OTC $0.9933 \pm .0023$. HyMeYOLO Ricci 5-seed backfill.
May 14	Niitsuma brief (+ PDF) shipped. Cliques foundation baseline. Gömb wiki datasets ($n=5$). Gömb-strict 4-dataset 5-seed validation; Epinions $0.9526 \pm .0018$ SOTA . Executive briefs Hu + En.
May 15	GömbSoma-RicciStim 10-phase + Cluttered MNIST training-infra complete; 4 optimisation passes give $296\times$ speedup (8283 ms $\rightarrow$ 28 ms). 150/150 tests pass. SDRF wiring optimisation.
May 16	Hodge real hot spot diagnosis (4E sparse.mm dominates); Rust quadtree port via PyO3 ($3.99.8\times$ speedup). HSiKAN tabular sanity null. HyMeYOLO honest-mAP baseline (the GT-consumption bug fix). HyMeYOLO Stage A-2 WIN ($0.7460 \pm .035$, $z = +14$). SDRF net-negative call on Cluttered MNIST.
May 17	2025 SOTA verbatim comparison: HSiKAN-Optuna +8.57 pp BA, +5.11 pp OTC vs DADSGNN. Stage B b_hsiKAN 4-seed tie. Stage C 5-seed report.
May 18	GZ + ROS 2 rapport demo shipped ; HyMeKo $\rightarrow$ SDF loop closed. IMDB arch-fairness. Stage D-1 falsified + diagnosed. Stage D-3 nodelet head + bis + tris + quater + quinquies (full series). Pimentel dossier (~ 50 -LOC bridge identified). IMDB
May 19 (today)	Stage P-mo multi-objective ABB with admissibility theorem + 9 tests. Methanol-synthesis worked example. Stage P-io bidirectional .pgip bridge in Rust with 5 tests. P-graph chapter-validation report. Relaxed-MSG knob + 4 tests. NN-architecture P-graph demo. Formalism-extension paper (Theorems 1 + 2). Csapó overview (this document) . P-engine Slice 1 (Rust builder, 9 tests).

18 Glossary and notation

Symbol / term	Meaning
M, O, R, P	Materials, operating units, raw set, product (demand) set in a P-graph
$\text{in}(u), \text{out}(u)$	A unit's input / output material sets
$C_R(O')$	Producibility closure: least fixed point of producibility from R under O'
$\mathfrak{S}(\mathcal{P})$	Set of feasible solution structures
$(\mathcal{P})$	Maximal structure: largest $O' \subseteq O$ that survives forward + backward trims
$\mathbf{c}(u)$	Multi-dimensional cost vector of unit u , $\mathbf{c}(u) \in \mathbb{R}_{\geq 0}^D$
$\mathbf{w}$	Weight vector $\mathbf{w} \in \mathbb{R}_{\geq 0}^D$ for multi-objective ABB
$\sigma(e)$	Sign of edge e , $\sigma(e) \in \{-1, +1\}$
$\prod_i \sigma(e_i)$	Cartwright-Harary balance of a signed cycle
α_k	Softmax mixing weight for cycle arity k in HSiKAN
$\phi(t)$	Catmull-Rom interpolant for a KAN-block activation
M_{CR}	Catmull-Rom basis matrix (4×4, from §3.3)
Δ_0	Vertex-Laplacian, used in GömbSoma's Hodge eigenmode pipeline
κ	Forman discrete Ricci curvature
ABB	Accelerated Branch-and-Bound (Friedler-Tarján-Huang-Fan 1992)
MSG	Maximal Structure Generation
SSG	Solution Structure Generation
HSiKAN	Hypergraph Signed Kolmogorov-Arnold Network (this work's signed-link model)
GömbSoma	Cortical-circuit hypergraph architecture; uses Forman κ , Hodge Δ_0 , Bochner conv
P-mo	Stage P-multi-objective (Theorem 1, 2026-05-19)
P-io	Stage P-io: bidirectional .pgip read/write (2026-05-19)
P-engine	Programmatic P-graph builder + PyO3 binding; Slice 1 shipped today

19 Code excerpts: the P-engine Slice 1 API

The smallest illustration of why a programmatic builder matters. HSiKAN architecture choice from Python is going to look like this (Slices 2-4 wrap a PyO3 binding around what we shipped tonight):

```
from hymeko import PGraphEngine

eng = PGraphEngine()
# Raw resource budgets and the validated-model demand.
eng.add_material("gpu_budget",      kind="raw")
eng.add_material("wall_budget",     kind="raw")
eng.add_material("cycle_pool")
eng.add_material("backbone_feats")
eng.add_material("trained_state")
eng.add_material("validated_model", kind="product")

# Cycle-enumeration choices (4 alternatives).
eng.add_unit(
    "cycle_k3_topm16",
    inputs=["gpu_budget", "wall_budget"],
    outputs=["cycle_pool"],
    cost=10.0,
    multi_cost={"gpu_gb": 0.4, "wall_min": 0.8, "auc_gap": 0.040},
)
eng.add_unit(
    "cycle_k345_topm64",
    inputs=["gpu_budget", "wall_budget"],
    outputs=["cycle_pool"],
    cost=80.0,
    multi_cost={"gpu_gb": 2.4, "wall_min": 6.0, "auc_gap": 0.010},
)

# ... three backbone widths, three training schedules, one eval ...

best = eng.solve_abb(weights={
    "gpu_gb": 1.0,
    "wall_min": 1.0,
    "auc_gap": 10000.0, # AUC-paramount regime
})
```

```
print(f"Optimal HSiKAN config: {best.units}, wcost = {best.cost}")
# Today's empirical answer (under this regime):
# {h32, cycle_k345_topm64, train_long_optuna_200ep, eval_5seed}
# wcost = 251.8
```

The Rust side (Slice 1, shipped tonight) is structurally identical:

```
use hymeko_pgraph::builder::{PgraphBuilder, MaterialKind};
use hymeko_pgraph::abb::{AbbOptions, solve_with_options};
use hymeko_pgraph::msg::maximal_structure;

let mut b = PgraphBuilder::new();
b.add_material("Toluene", MaterialKind::Raw)?;
b.add_material("H2", MaterialKind::Raw)?;
b.add_material("Mix", MaterialKind::Intermediate)?;
b.add_material("Benzene", MaterialKind::Product)?;
b.add_material("Methane", MaterialKind::Intermediate)?;
b.add_unit("Mixer", &["Toluene", "H2"], &["Mix"], 100.0)?;
b.add_unit("Reactor", &["Mix"], &["Benzene", "Methane"], 250.0)?;
b.add_unit("DirectSynth", &["Toluene", "H2"], &["Benzene"], 800.0)?;
b.add_unit("Disposal", &["Methane"], &[], 50.0)?;

let graph = b.build()?;
let msg = maximal_structure(&graph);
let sol = solve_with_options(&graph, &msg, AbbOptions::default());
// sol.units == {Mixer, Reactor, Disposal}, sol.cost == 400.0
```

9 builder tests pass on first run:

Test	Status
empty_builder_build_fails_no_materials	✓
builder_with_materials_but_no_units_fails	✓
duplicate_material_name_fails	✓
unit_name_collides_with_material_name_fails	✓
unit_referencing_unknown_material_fails_at_build_time	✓
build_hda_reproduces_textbook_abb_optimum	✓
multi_cost_dimensions_alphabetised_at_build_time	✓
multi_cost_with_missing_dim_pads_to_zero	✓
programmatic_hda_matches_hymeko_lowered_hda	✓

20 What the next 19 days should do

Priority	Item	Why
1	Cortical benchmark (Cichy 92 + Brain-Score)	Right falsification for GömbSoma
2	5-seed Stage H VOC	Paper-citation quality for the visit demo
3	P-engine Slice 2 (PyO3 binding)	Unblocks HSiKAN/Gömb in-process use
4	P-engine Slice 3 (Python wrapper)	Brings the engine to PyTorch users
5	Strict-protocol σ -masked cycles fix	Closes Epinions 0.9526 interpretation gap
6	HyMeYOLO architecture in HyMeKo (full VOC)	“one DSL, one substrate” completion
7	Stage B' retry (concurrent-GPU)	Owed since May 17
8	MEMORY.md trim (31 kB $\rightarrow$ < 24 kB)	CLAUDE.md system-reminder load

For Csapó specifically

Three places your perspective would change this most:

1. **The HSMM compiler thread** (Nagare $\rightarrow$ HSMM, frozen since 2026-05-11) currently parked behind an FPGA prerequisite gate. Does the 19-day work change the reopen-criteria, or should the gate stand as written? The multi-objective ABB (Theorem 1 above) might let the gate become a parametric one rather than a binary go/no-go: we could express the FPGA-vs-CPU-vs-GPU choice as a multi-cost ABB problem on the compiler-toolchain P-graph, and let the weighted optimum say whether to reopen.
2. **The cortical benchmark falsification framing.** The GömbSoma architecture is built to be a cortical-circuit model; Cluttered MNIST tested the wrong reflex. Do you have a view on whether Cichy 92 is the

right “first viable” target, or whether we should aim directly at NSD shared-1000 (8 subjects, 1000 images, fMRI)? Both have public Brain-Score bindings; Cichy is smaller and ships faster; NSD is more authoritative.

3. **The joint-paper framing.** The P-graph methodology extension is, in spirit, a Hungarian methodology coming home: Friedler’s 1992 ABB is from Veszprém; our 2026 extension to neural-architecture and signed-graph cycle enumeration is from the same intellectual lineage. Is there value in framing the joint paper as a “Hungarian methodology beyond PSE” contribution, or should we keep the framing strictly algorithmic? The former is more visible; the latter is more defensible at peer review.

Reading order for the companion artefacts

If Csapó wants to read more, here is the suggested order:

1. This document (overview).
2. `reports/2026-05-19-pgraph-formalism-extended.pdf` the 7-page formal paper with Theorems 1 + 2.
3. `reports/2026-05-19-pgraph-multi-objective.pdf` methanol-synthesis worked example (5 pages, `.tex` + `.pdf`).
4. `reports/2026-05-19-pgraph-nn-architecture-search.pdf` HSiKAN as a P-graph (4 pages).
5. `reports/2026-05-18-pimentel-abb-ssg-msg-summary.pdf` the survey of where the P-graph machinery stood pre-extension (7 pages).
6. `reports/2026-05-19-pgraph-pgip-roundtrip.md` the bidirectional `.pgip` bridge (Markdown only; companion to #5).
7. `reports/2026-05-19-pgip-chapters-validation.md` the chapter validation report; the strict-vs-relaxed MSG finding lives here.
8. `reports/2026-05-14-niitsuma-brief.pdf` the visit-target brief from one week ago.
9. Source: `hymeko_pgraph/src/` for the algorithm implementations; `signedkan_wip/src/hymeko_gomb/soma/` for GömbSoma.

Closing

The 19 days were structured (after May 10) around the operating contract; the contract did most of the heavy lifting. Every change has a plan, tests, a report, and (for paper-headline claims) an n -seed validation. The negative-results catalogue is as well-documented as the wins.

The framework’s intellectual lineage signed-graph theory, P-graph process synthesis, Kolmogorov-Arnold representation is clearer now than it was on May 1. The unification of cycle enumeration and chemical-process synthesis under one abstract algorithm (Theorem 2) is the most “surprising” theoretical result of the cycle. Friedler’s 1992 framework reaches further than its authors stated. So does Heider’s 1946 balance.

HYMEKO is at the point where every artefact you would want to hand to a collaborator (PDF brief, code path, reproducible run, mathematical proof, statistical validation) is on disk and runs. The next 19 days should be about deploying that machinery on the two biggest open targets: the cortical benchmark for GömbSoma and the in-process PyTorch layer for the P-engine.

Acknowledgements. The 19 days drew on the Friedler-Tarján- Huang-Fan 1992 ABB, the Cartwright-Harary 1956 generalised balance theorem, the Heider 1946 triadic balance, the Liu et al. 2024 KAN construction, the Catmull-Rom 1974 spline, and the rapidly-evolving 2025 signed-graph literature (SE-SGformer, DADSGNN). The Triton fused-kernel pattern was inspired by the FlashAttention work (Dao et al.).

This document was assembled from the project memory (113 entries), ~ 75 reports, 23 plan dirs, and the git log across the `refactor/extract-hymeko-hre` and `master` branches.